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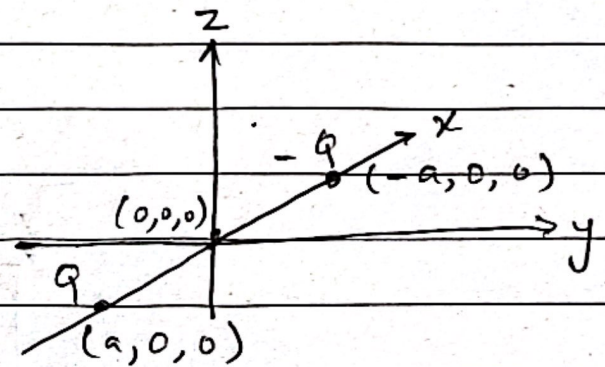
Q.1) A point Q is located at $(a, 0, 0)$, while another charge $-Q$ is at $(-a, 0, 0)$. Find the electric field E at the following field points.

(a) $(0, 0, 0)$

$$\Rightarrow E = kQ \left(\frac{\vec{R}_+}{|\vec{R}_+|^3} - \frac{\vec{R}_-}{|\vec{R}_-|^3} \right)$$

$$\vec{R}_+ = r - \langle a, 0, 0 \rangle$$

$$\vec{R}_- = r - \langle -a, 0, 0 \rangle$$



$$\vec{R}_+ = \langle 0, 0, 0 \rangle - \langle a, 0, 0 \rangle = \langle -a, 0, 0 \rangle$$

$$\vec{R}_- = \langle 0, 0, 0 \rangle - \langle -a, 0, 0 \rangle = \langle a, 0, 0 \rangle$$

$$|\vec{R}_+| = \sqrt{(-a)^2 + 0^2 + 0^2} = a$$

$$|\vec{R}_-| = \text{ " " " } = a$$

$$E = kQ \left(\frac{\langle -a, 0, 0 \rangle}{a^3} - \frac{\langle a, 0, 0 \rangle}{a^3} \right) = kQ \left(\frac{-2a}{a^3} \right)$$

$$E = \frac{-2kQ}{a^2}$$

Date: _____

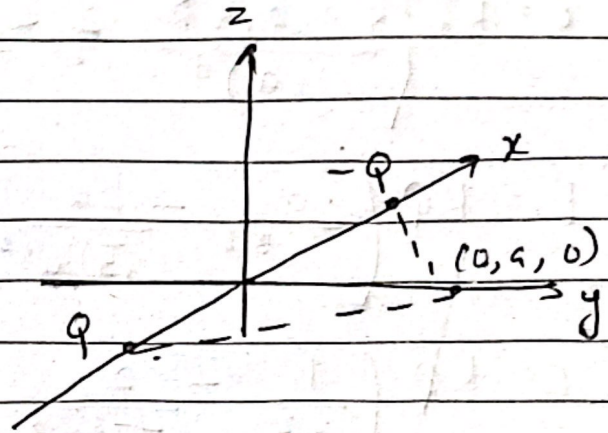
b) $(0, a, 0)$

$$\vec{r}_+ = \langle 0, a, 0 \rangle - \langle a, 0, 0 \rangle$$

$$\vec{r}_- = \langle 0, a, 0 \rangle - \langle -a, 0, 0 \rangle$$

$$\vec{r}_+ = \langle -a, a, 0 \rangle$$

$$\vec{r}_- = \langle a, a, 0 \rangle$$



$$|\vec{r}_+| = \sqrt{(-a)^2 + (a)^2 + 0} = \sqrt{2} a$$

$$|\vec{r}_-| = \sqrt{a^2 + a^2 + 0} = \sqrt{2} a$$

$$E = kq \left(\frac{\langle -a, a, 0 \rangle}{(\sqrt{2} a)^3} - \frac{\langle a, a, 0 \rangle}{(\sqrt{2} a)^3} \right) = kq \left(\frac{-2a}{(\sqrt{2} a)^3} \right)$$

$$E = kq \left(\frac{1a}{\sqrt{2} a^2} \right) = \boxed{\frac{kq}{a^2 \sqrt{2}}}$$

c) $(a, 0, a)$

$$\vec{r}_+ = \langle a, 0, a \rangle - \langle a, 0, 0 \rangle$$

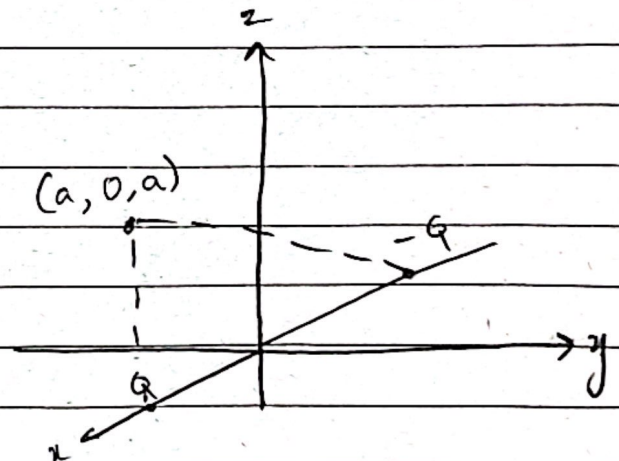
$$= \langle 0, 0, a \rangle$$

$$\vec{r}_- = \langle a, 0, a \rangle - \langle -a, 0, 0 \rangle$$

$$= \langle 2a, 0, a \rangle$$

$$|\vec{r}_+| = \sqrt{(0)^2 + (0)^2 + (a)^2} = a$$

$$|\vec{r}_-| = \sqrt{(2a)^2 + (0)^2 + (a)^2} = \sqrt{5} a$$



Date: _____

$$E = kQ \left(\frac{\langle 0, 0, a \rangle}{(a)^3} - \frac{\langle 2a, 0, a \rangle}{(\sqrt{5}a)^3} \right)$$

$$E = kQ \left(\frac{a}{a^{3/2}} - \frac{2a}{(\sqrt{5})^3 a^{3/2}} - \frac{a}{(\sqrt{5})^3 a^{3/2}} \right)$$

$$E = kQ \left(\frac{1}{a^2} - \frac{2}{\sqrt{5}^3 a^2} - \frac{1}{\sqrt{5}^3 a^2} \right)$$

$$E = kQ \left(\frac{5^{3/2} - 2 - 1}{(\sqrt{5})^3 a^2} \right)$$